

Thomas Constertina

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Education

The University of Texas at Austin (UT-Austin)

- **Master of Science in Artificial Intelligence (January 2024 – December 2024)**
 - GPA: 3.63/4.0
 - Focus areas included generative AI, automated logical reasoning, natural language processing, machine learning, reinforcement learning, search and reasoning under uncertainty, optimization, and more.

University of Illinois at Urbana-Champaign (UIUC)

- **Master of Science in Computer Science (January 2023 – August 2023)**
 - Graduate GPA: 3.87/4.0
 - Focus areas included artificial intelligence, cloud computing, advanced algorithms, data structures, and other technologies to facilitate machine learning on large, complex data sets.
- **Bachelor of Arts in Computer Science & Chemistry (August 2019 – December 2022)**
 - Undergraduate GPA: 3.78/4.0
 - UIUC Computational Chemistry Researcher
 - R Programming Club
 - Achieved perfect college entrance exam score (ACT)

Technical Skills

- Language proficiency with C++, Python, Java, Assembly, C, C#, SQL, R, Bash, Golang, and Javascript. Python AI libraries used include TensorFlow, PyTorch, Numpy, SciPy, Keras, and Pandas.
- Cloud-based development proficiency leveraging AWS clusters, Lambda functions, EC2 instances, microservices, cloud storage, databases, API development, caching, etc.
- Front-end web application development proficiency using Angular, React, HTML/CSS, JavaScript, and Vue.

Computer Science Coursework

- **Undergraduate Computer Science Coursework:** Computer Science I, Computer Science II, Software Design Lab, Data Structures, Discrete Structures, Computer Architecture, Algorithms, System Programming
- **Graduate Level Computer Science Coursework:** Database Systems, Web Programming, Software Engineering, Cloud Networking, Computer Security, Cloud Computing Applications, Deep Learning, Programming Languages & Compilers, Data Mining, Internet of Things, Theory & Practice of Data Cleaning, Scientific Visualization, Ethics in AI, Online Learning & Optimization, Automated Logical Reasoning, Machine Learning

Employment

- Software Developer (Contract), *SigLens* (January 2025 – February 2025)
 - Implemented role-based access control (RBAC) to manage user and group-level permissions for data access
 - Built core functionality for a parser to support a new query language
 - Made comprehensive end-to-end test cases on all new features
- Generative AI Engineer (Contract), *Scale AI* (November 2023 – November 2024)
 - Trained generative AI coding models by solving complex coding problems in C++
 - Interviewed programmers for software engineering roles
- Software Engineer Intern, *Motorola Solutions* (June 2022 – August 2022)
 - Developed C++ firmware for RF channel hardware devices used by first responders
 - Created an alarm-handling system in C++ to monitor radio hardware status
 - Practical experience working with docker containers and Linux virtual machines
 - Engaged throughout entire SDLC including planning, design, development, and testing
 - Worked on a scrum team leveraging Agile methodology

- Computational Chemistry Researcher, *The Moore Group, University of Illinois* (June 2020 - August 2021)
 - Leveraged technology and scientific skills using Python to perform Data Analysis of Molecular Systems
 - One of only four undergraduates working with graduate and postdoctoral researchers
 - Works: “Using a Rule-Based Model to Simulate Strand Escape from an [n]-Rung Ladder, Accounting for Diagonal Bond Formation and Rotation”

Certifications

- AWS Certified Cloud Practitioner, 2023